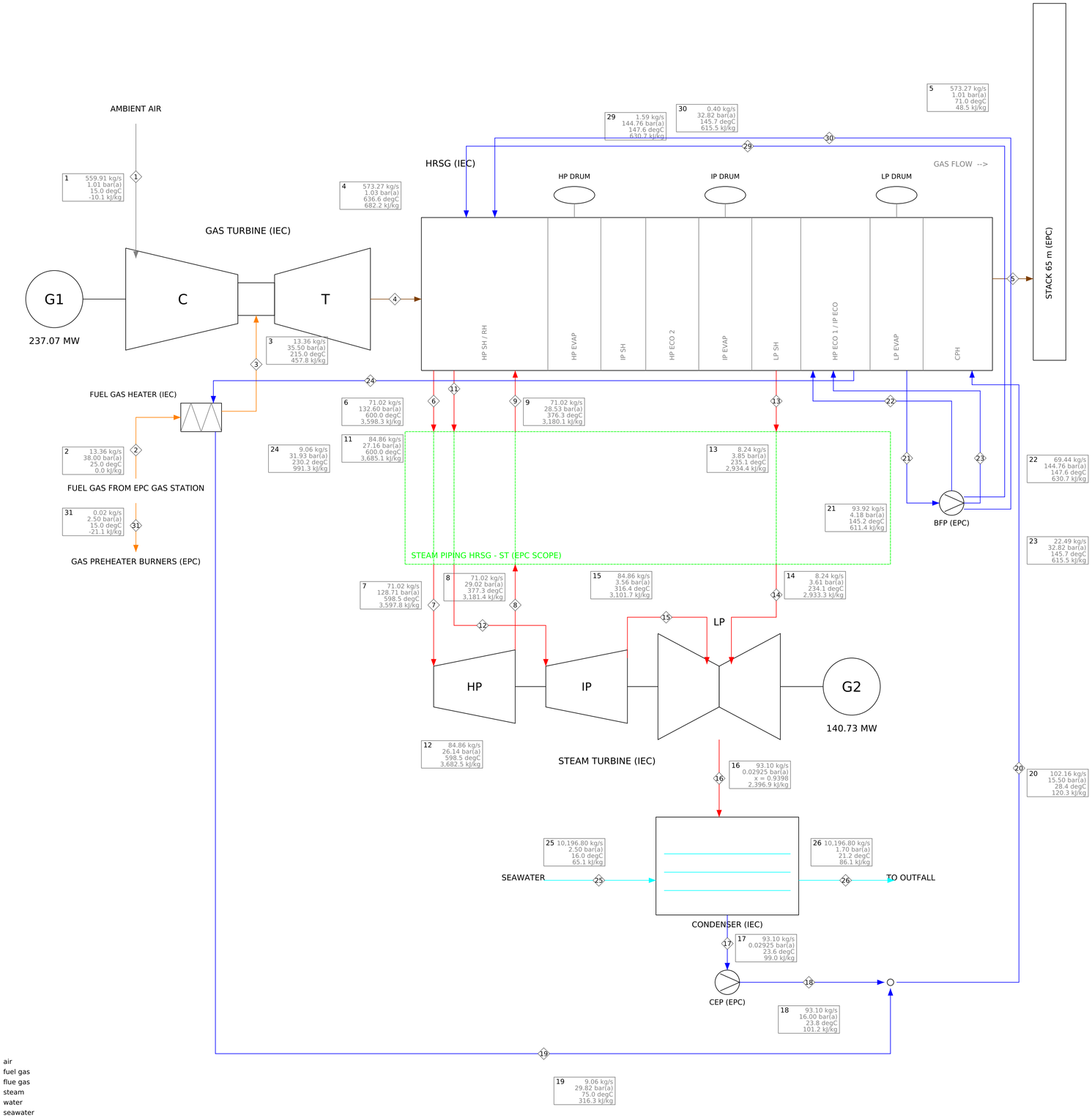


STREAM TABLE - SRC-NG-060

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	559.91	1.01	15.0	-10.1	-5.7
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	13.36	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	13.36	35.50	215.0	457.8	6.1
4	GT exhaust gas to HRSG inlet	573.27	1.03	636.6	682.2	391.1
5	Flue gas at HRSG outlet to stack	573.27	1.01	71.0	48.5	27.8
6	HP steam at HRSG outlet (HP SH outlet)	71.02	132.60	600.0	3,598.3	255.6
7	HP steam at ST main stop valve inlet	71.02	128.71	598.5	3,597.8	255.5
8	Cold reheat steam at HP turbine exhaust	71.02	29.02	377.3	3,181.4	226.0
9	Cold reheat steam at HRSG reheater inlet	71.02	28.53	376.3	3,180.1	225.9
10	IP superheated steam (joins cold reheat inside the HRSG)	13.43	29.19	310.9	3,024.1	40.6
11	Hot reheat steam at HRSG outlet	84.86	27.16	600.0	3,685.1	312.7
12	Hot reheat steam at ST IP stop/control valve inlet	84.86	26.14	598.5	3,682.5	312.5
13	LP steam at HRSG outlet	8.24	3.85	235.1	2,934.4	24.2
14	LP steam at ST LP admission	8.24	3.61	234.1	2,933.3	24.2
15	IP turbine exhaust (crossover, internal)	84.86	3.56	316.4	3,101.7	263.2
16	LP turbine exhaust to condenser (UEEP)	93.10	0.0293	x 0.940	2,396.9	223.1
17	Condensate at hotwell outlet / CEP suction	93.10	0.0293	23.6	99.0	9.2
18	Condensate at CEP discharge	93.10	16.00	23.8	101.2	9.4
19	Fuel gas heater water return to condensate	9.06	29.82	75.0	316.3	2.9
20	Condensate to HRSG condensate preheater inlet	102.16	15.50	28.4	120.3	12.3
21	Feedwater from LP drum to boiler feed pumps	93.92	4.18	145.2	611.4	57.4
22	HP feedwater at BFP discharge to HP economiser	69.44	144.76	147.6	630.7	43.8
23	IP feedwater at BFP interstage to IP economiser	22.49	32.82	145.7	615.5	13.8
24	IP economiser water to performance fuel gas heater	9.06	31.93	230.2	991.3	9.0
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	21.2	86.1	878.0
29	HP attemperation spray water (BFP discharge)	1.59	144.76	147.6	630.7	1.0
30	Reheat attemperation spray water (BFP interstage)	0.40	32.82	145.7	615.5	0.2
31	Fuel gas to the station preheater burners (water-bath heaters)	0.02	2.50	15.0	-21.1	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	237.07	MW
ST output (generator terminals)	140.73	MW
Gross output	377.80	MW
Auxiliary loads	-8.20	MW
Transformer losses (GSU, UAT, MV/LV)	-1.06	MW
NET OUTPUT (HV terminals)	368.54	MW
Heat input (LHV)	635.13	MW
Gross heat rate (LHV)	6,043	kJ/kWh
NET HEAT RATE (LHV)	6,204	kJ/kWh
Net efficiency (LHV)	58.03	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	65,741	Sm3/h
Circulating water flow	36,112	m3/h
GT load	55.78	%
PLANT LOAD (NET, % OF SRC-NG-100)	60.03	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.004 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.184 MW (tolerance 0.89 MW)
OK	energy balance FGH: residual +0.002 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.013 MW (tolerance 0.63 MW)
OK	energy balance HRSG: residual +1.482 MW vs stated casing loss 1.453 MW (tolerance 0.69 MW)
OK	energy balance PIPE: residual +0.362 MW (tolerance 0.82 MW)
OK	energy balance ST: residual -0.002 MW (tolerance 0.59 MW)
OK	CW temperature rise 5.16 K vs DP-CASE-0015 <= 7.0 K (Condenser cooling water temperature rise)
OK	gross heat rate 6,042.65 kJ/kWh vs IG-07 <= 6,060.00 (margin +0.29 %)
OK	net heat rate 6,204.01 kJ/kWh vs PG-03 <= 6,250.00 (margin +0.74 %)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-060
CASE	Site Reference Conditions, natural gas, 60 % plant net load (guarantee PG-03)
DOCUMENT	ALP-HMB-SRC-NG-060
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	6b60fce553593bf5 \| 2026-09-29
NOTE	all values calculated from the database; do not edit